The object classifier square of a function is a pullback

The Univalent Foundations Program

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Editorial note. The counted statement is the author's 'thm:object-classifier' (source lines 859-874, Theorem 4.8.4); the proof environment starts on the next line, so the pairing is by adjacency.

Editorial note. The retained counter values (chapter 4, section 8, theorem offset 3) make the counted theorem print as Theorem 4.8.4, matching the original book numbering taken from the label-number dump of the pinned snapshot.

Editorial note. No context span is retained: the statement and proof are self-contained apart from one labelled invocation.

Editorial note. This source directory intentionally contains only the author file, so the retained book macro library is bundled as a per-task copy inside this task directory and referenced from preamble_spans; the copy is verbatim from the same pinned book snapshot and no macro is added, changed or invented.

Editorial scope. One original result of the HoTT Book, counted once: the object classifier square is a pullback (Theorem 4.8.4).

Allowed standard prerequisites (editorial).

- Pullback square. A commutative square is a pullback when the induced map from the domain into the pullback of the cospan is an equivalence.. It is the conclusion of the theorem, in the sense of the retained exercise.
- Pointed types. pointed type is the type of pairs (X, x) with x : X, and proj1 is the forgetful map to type.. It is the codomain of the map vartheta_f whose square is analysed.
- Univalence. An equivalence of types gives a path between them, so an equivalence between A and a pullback gives the required identification.. It underlies the chain of equivalences exhibited in the proof.

Source reference. Original exercise 2.11: sources/hott-hits/book-labelnumbers-first-edition.tex, source lines 1389–1389. Original result cited by the retained text but located outside the retained span. It is a navigation pointer only: the counted statement's own author proof is retained line for line and no part of it is replaced by this reference. The reference number is taken from the label-number dump of the pinned book snapshot.

Theorem 4.8.4. *Let $f : A \to B$ be a function. Then the diagram*

$$\begin{array}{ccc} A & \xrightarrow{\vartheta_f} & \mathcal{U}_\bullet \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle \mathsf{pr}_1} \\ B & \xrightarrow[\chi_f]{} & \mathcal{U} \end{array}$$

is a pullback square (see original exercise 2.11). Here the function ϑ_f is defined by

$$\lambda a.\, (\mathsf{fib}_f(f(a)), (a, \mathsf{refl}_{f(a)})).$$

Proof. Note that we have the equivalences

$$\begin{aligned} A &\simeq \sum_{b:B} \mathsf{fib}_f(b) \\ &\simeq \sum_{(b:B)} \sum_{(X:\mathcal{U})} \sum_{(p:\mathsf{fib}_f(b)=X)} X \\ &\simeq \sum_{(b:B)} \sum_{(X:\mathcal{U})} \sum_{(x:X)} \mathsf{fib}_f(b) = X \\ &\simeq \sum_{(b:B)} \sum_{(Y:\mathcal{U}_\bullet)} \mathsf{fib}_f(b) = \mathsf{pr}_1 Y \\ &\equiv B \times_{\mathcal{U}} \mathcal{U}_\bullet \end{aligned}$$

which gives us a composite equivalence $e : A \simeq B \times_{\mathcal{U}} \mathcal{U}_\bullet$. We may display the action of this composite equivalence step by step by

$$\begin{aligned} a &\mapsto (f(a),\ (a, \mathsf{refl}_{f(a)})) \\ &\mapsto (f(a),\ \mathsf{fib}_f(f(a)),\ \mathsf{refl}_{\mathsf{fib}_f(f(a))},\ (a, \mathsf{refl}_{f(a)})) \\ &\mapsto (f(a),\ \mathsf{fib}_f(f(a)),\ (a, \mathsf{refl}_{f(a)}),\ \mathsf{refl}_{\mathsf{fib}_f(f(a))}) \\ &\mapsto (f(a),\ (\mathsf{fib}_f(f(a)),\ (a, \mathsf{refl}_{f(a)})),\ \mathsf{refl}_{\mathsf{fib}_f(f(a))}). \end{aligned}$$

Therefore, we get homotopies $f \sim \mathsf{pr}_1 \circ e$ and $\vartheta_f \sim \mathsf{pr}_2 \circ e$. □